



Bluetooth

Session 17

dot
LEBANON
Activate Windows
Go to Settings to activate Windows

mint mobile.com is sharing your screen

Stop sharing

Hide

Objectives

Connect the Bluetooth to Arduino Board

Use the Bluetooth features

Use the android application to communicate with Arduino

Control LED using the Bluetooth connection.

Bluetooth

Activate Windows
Go to Settings to activate Windows.

Bluetooth

An Arduino Bluetooth board is a small electrical chip.

The Bluetooth board has some major benefits like:

1. Allows the transfer of sensor values to your laptop or Phone
2. In the case of robotics, it allows the control of the robot directly for phone applications.
3. In case of multiple Arduino boards, it synchronizes them together

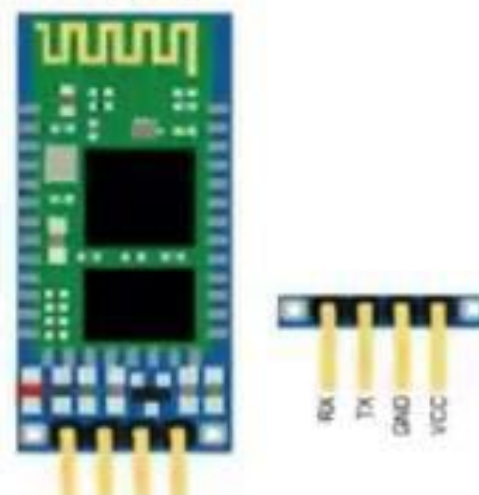


Bluetooth Pins

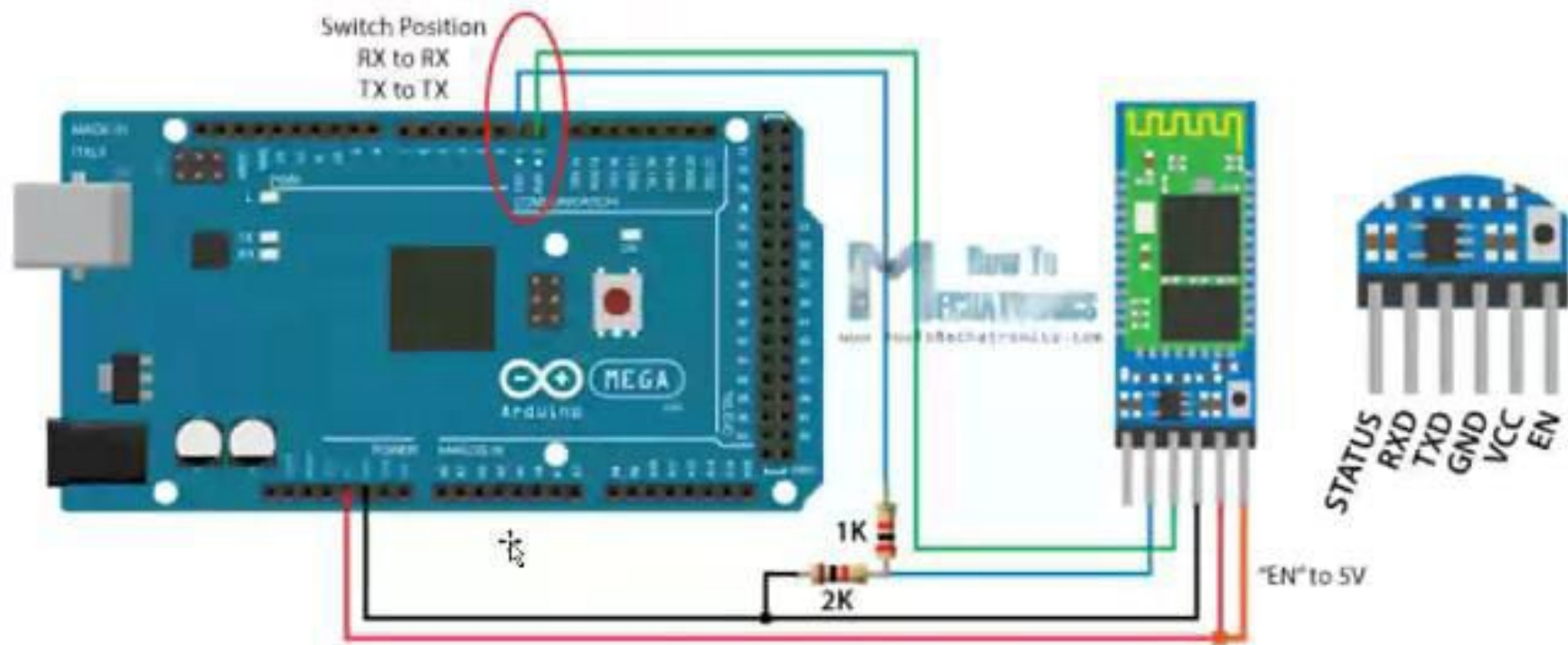
The Bluetooth (HC-06) has four pins.

- The VCC and GND are for power supply,
- The TX and RX are for sending data.
- The RX of the Bluetooth Board should be connected to the TX on the Arduino while the TX should be connected to the RX. The Arduino uses these pins to communicate with the computer, so when uploading the program from the computer to the Arduino both terminals should be disconnected.

HC-06

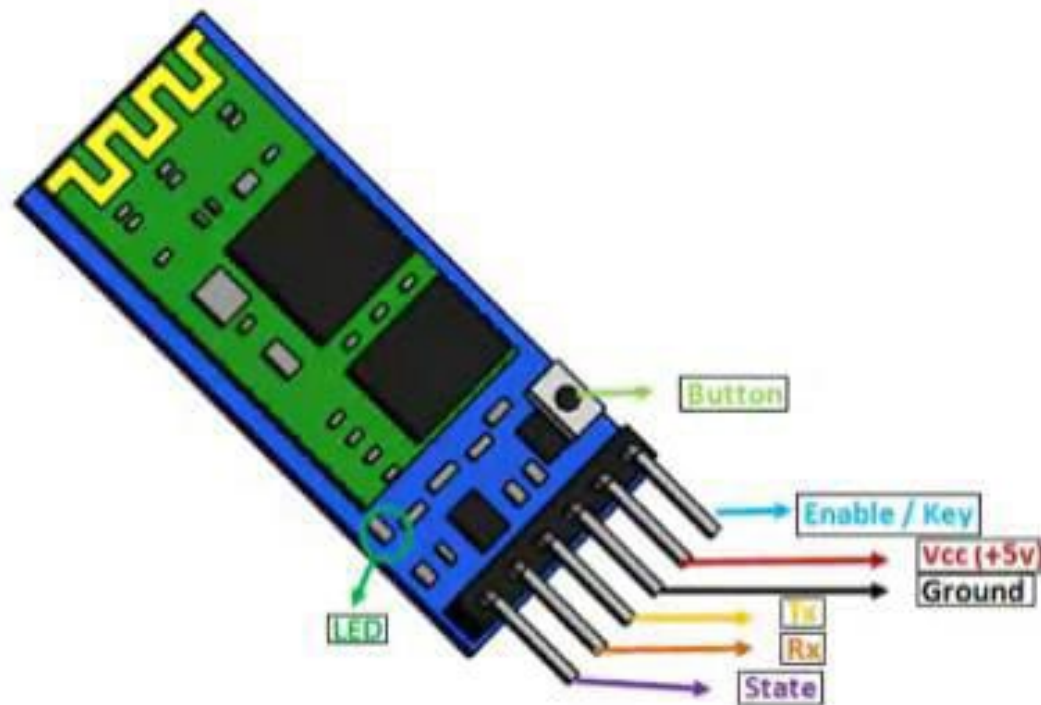


Board connection



Bluetooth Pins

The Bluetooth (HC-06) pins.



Bluetooth Connection



Once the Bluetooth is connected to the Arduino, it becomes a part of the Arduino Board. This means that in the program, it is not required to specify that this data should be received by the Bluetooth. This is because the Bluetooth and the serial monitor are connected together and the data send to the Bluetooth are read by the Arduino as received from the serial monitor.

Bluetooth Controller App

The following program has the basics required to control Arduino elements through Bluetooth. Write down the program and let's test it.



Bluetooth Controller App





LED/Lamp



RGB Picker



Car Controller



Switches



Gamepad



Terminal



Voice Terminal



4x4 Keypad

Activity: Control the LED level through Bluetooth

Bluetooth Controller App

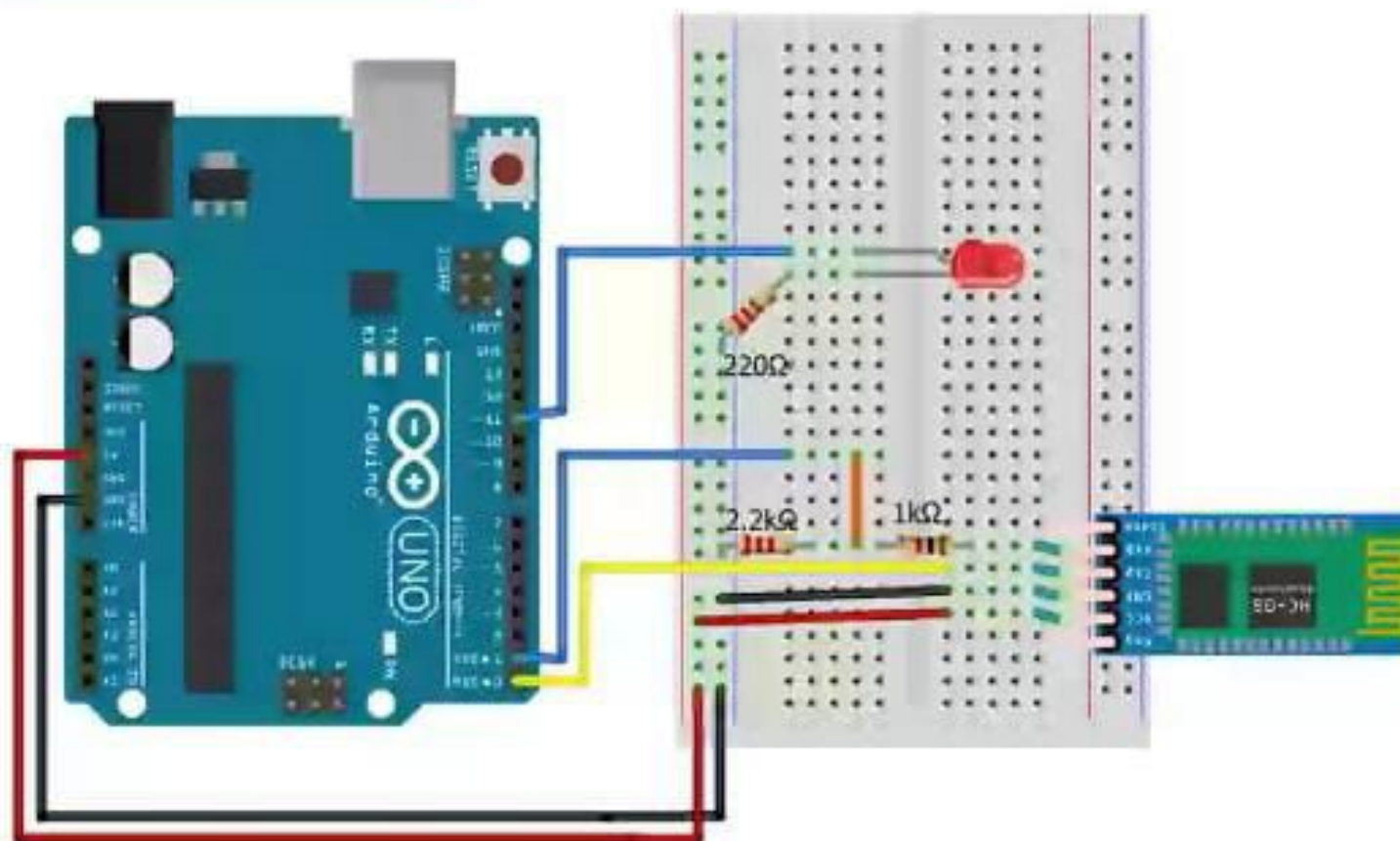


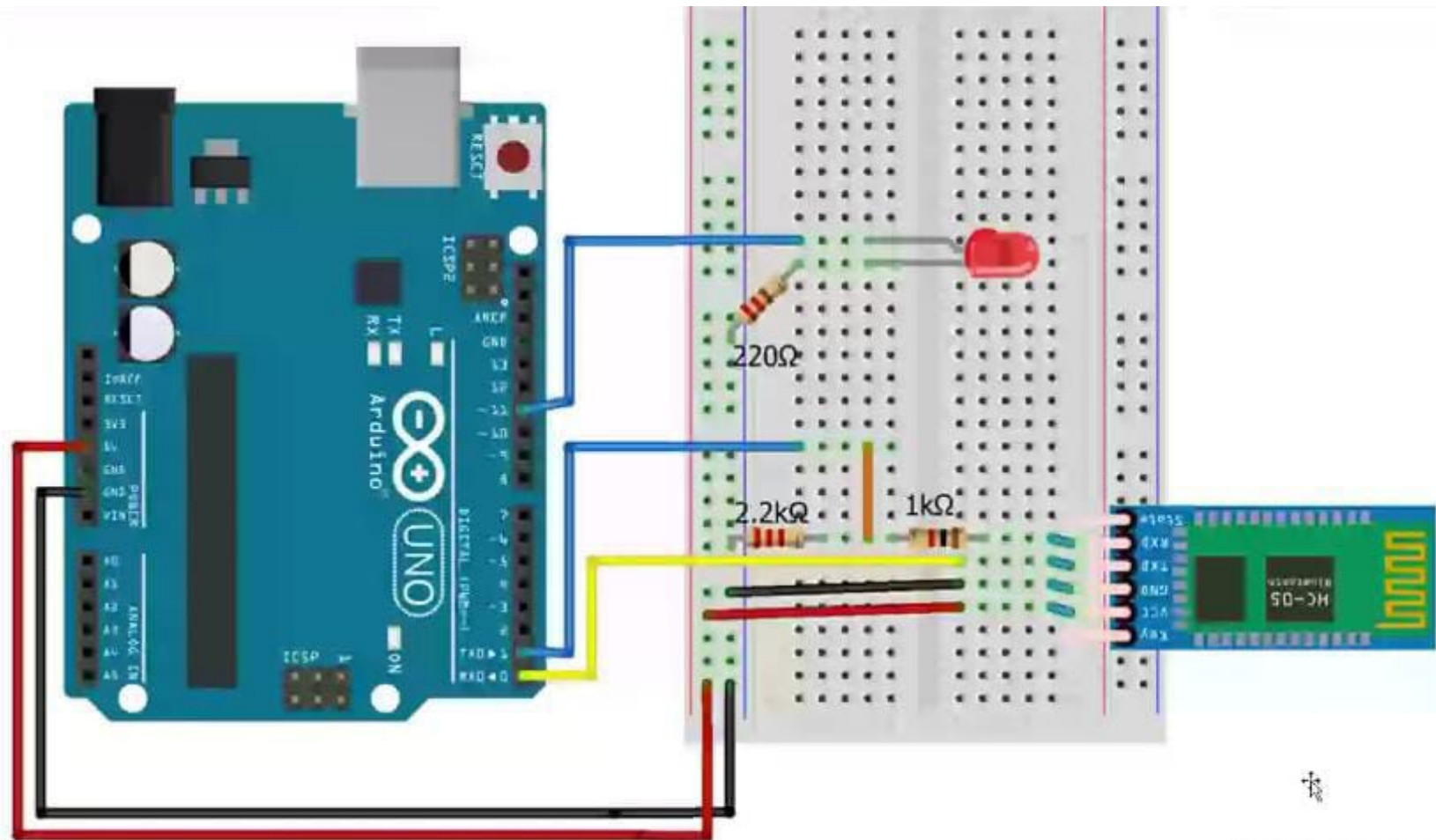
Potentiometer & serial monitor screen

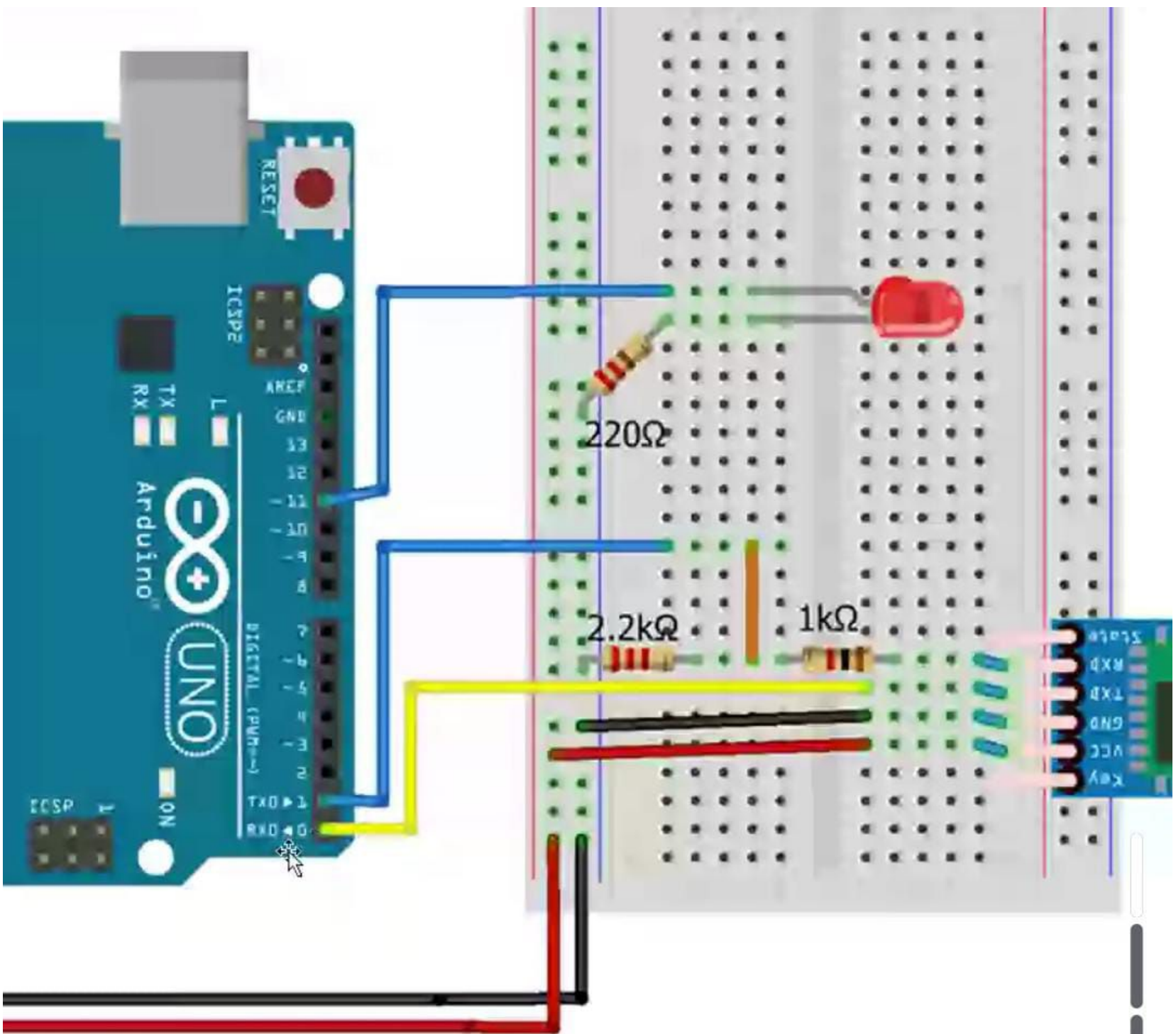
Required Elements:

1. Arduino Board
2. Bluetooth Board HC-05
3. Led
4. Resistors: 1K, 2.2 K, and 220 ohm
5. Breadboard

Board connection







Arduino Bluetooth Control



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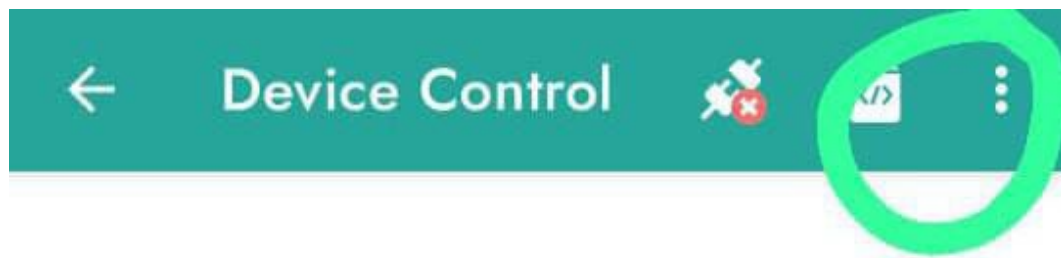
Terminal



Voice Terminal



4x4 Keypad



Change setting to 0 and
F



```
1 int led=3; //the led shloud be connected on a PWM pin
2 char inChar; //char type variable to save the data/readings/signals from the bluetooth
3 int intensity; //variable to control the light intensity using PWM signals
4
5 void setup() {
6   // put your setup code here, to run once:
7   pinMode(led,OUTPUT);
8   Serial.begin(9600);
9 }
10
11 void loop() {
12   // put your main code here, to run repeatedly:
13   if(Serial.available()){ //if receiving the data from the bluetooth device
14     inChar=Serial.read(); //Serial.read : command to read and save the data from the bluetooth module
15     Serial.println(inChar);
16   }
17   if(inChar=='0'){
18     analogWrite(led,
19       analogWrite(uint8_t pin,
20         int val) -> void
21       )
22   }
```



LED/Lamp



RGB Picker



Car Controller



Switches



Gamepad



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Voice Terminal



4x4 Keypad



Settings

Press

O

Release

Command to send (default: 0)

Down

Press

F

Release

Command to send (default: 0)

Right

Press

I

Release

Command to send (default: 0)

Left

Press

D



LED/Lamp



RGB Picker



Car Controller



Switches



Gamepad



Terminal



Voice Terminal



4x4 Keypad



Settings

Press

O

Release

Command to send (default: 0)

Down

Press

F

Release

Command to send (default: 0)

Right

Press

I

Release

Command to send (default: 0)

Left

Press

D



HC-05



1

Failed to send

1

05:33 PM

green led turned

05:33 PM

ON and the door is opened

05:33 PM

2

05:33 PM

red led turned ON and the door i

05:33 PM

s closed

05:33 PM

Type your command





HC-05



05:40 PM

red led turned ON and the door i

05:40 PM

s closed

05:40 PM

1

05:40 PM

green led turned ON and the door is opened

05:40 PM

2

05:40 PM

red led turned ON and the door is closed

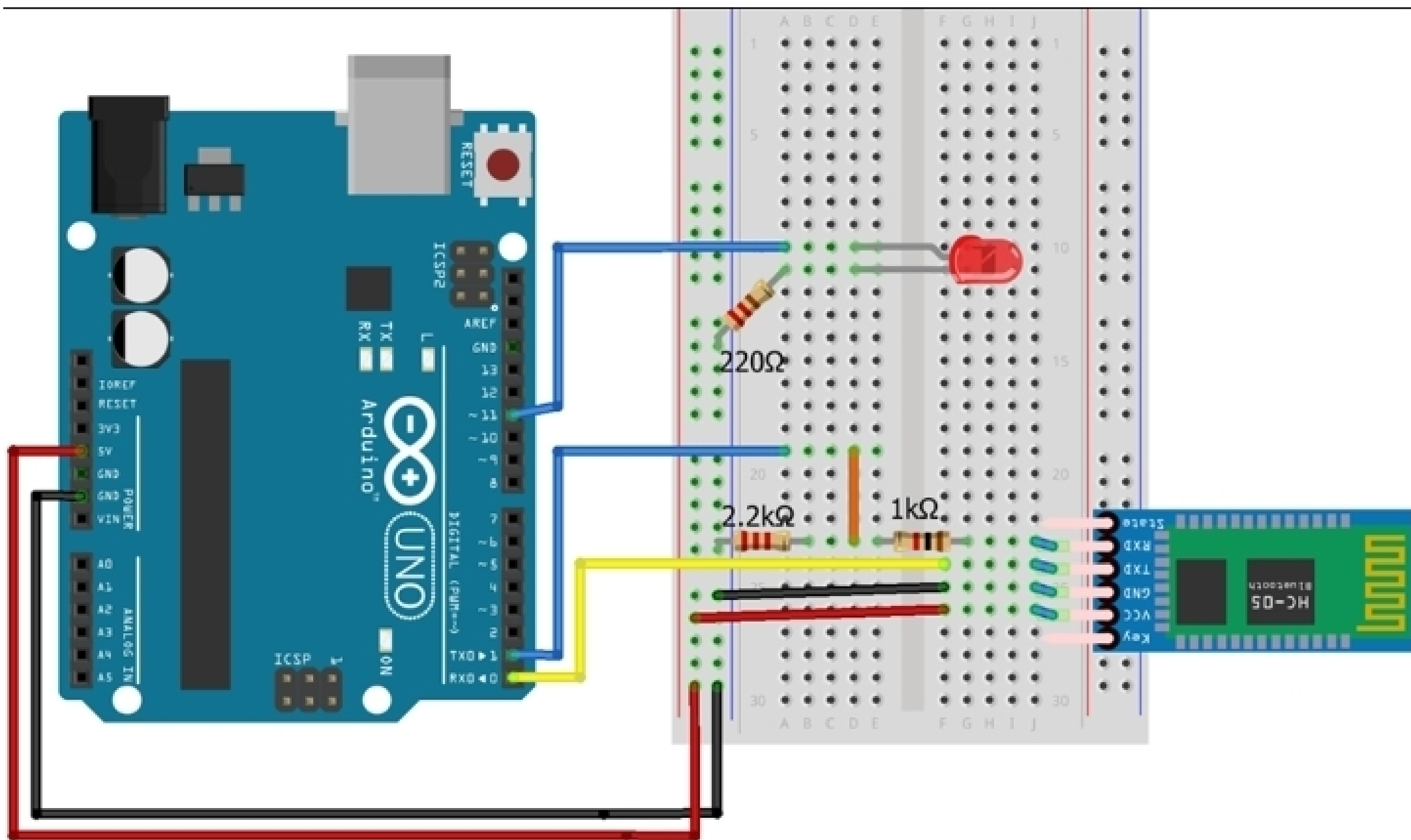
05:40 PM



05:40 PM

Type your command





Arduino Bluetooth Control Code

1. Basic LED On/Off via Bluetooth

```
int led=3;
char inChar; //char type variable to save the data/readings/signals from the bluetooth

void setup() {
    // put your setup code here, to run once:
    pinMode(led,OUTPUT);
    Serial.begin(9600);
}

void loop() {
    // put your main code here, to run repeatedly:
    if(Serial.available()){ //if receiving the data from the bluetooth device
        inChar=Serial.read(); //Serial.read : command to read and save the data from the bluetooth module
        Serial.println(inChar);
    }
    if(inChar=='0'){
        digitalWrite(led,HIGH);
    }else if(inChar=='F'){
        digitalWrite(led,LOW);
    }
}
```

2. LED Brightness Control (PWM) - v1

```
int led=3; //the led should be connected on a PWM pin
char inChar; //char type variable to save the data/readings/signals from the bluetooth
int intensity=25; //variable to control the light intensity using PWM signals

void setup() {
  // put your setup code here, to run once:
  pinMode(led,OUTPUT);
  Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
  if(Serial.available()){ //if receiving the data from the bluetooth device
    inChar=Serial.read(); //Serial.read : command to read and save the data from the bluetooth module
    Serial.println(inChar);
  }
  if(inChar=='0'){
    analogWrite(led,255); //turn on the led with a full brightness
  }
  else if(inChar=='F'){
    analogWrite(led,0); // turn off the led
  }
  else if(inChar=='I'){ //command to increase the light intensity
    intensity+=25; //increasing the light intensity of 25
    if(intensity>255){
      intensity=255;
    }
    analogWrite(led,intensity);
  }
  else if(inChar=='D'){ //command to decrease the light intensity
    intensity-=25;
    if (intensity<5){
      intensity=25;
    }
    analogWrite(led,intensity);
  }
}
```

3. LED Brightness Control (PWM) - v2

```
int led=3; //the led should be connected on a PWM pin
char inChar; //char type variable to save the data/readings/signals from the bluetooth
int intensity=50; //variable to control the light intensity using PWM signals

void setup() {
  // put your setup code here, to run once:
  pinMode(led,OUTPUT);
  Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
  if(Serial.available()){ //if receiving the data from the bluetooth device
    inChar=Serial.read(); //Serial.read : command to read and save the data from the bluetooth module
    Serial.println(inChar);
  }
  if(inChar=='0'){
    analogWrite(led,255); //turn on the led with a full brightness
  }
  else if(inChar=='F'){
    analogWrite(led,0); // turn off the led
  }
  else if(inChar=='I'){ //command to increase the light intensity
    intensity+=25; //increasing the light intensity of 25
    if(intensity>255) intensity=255;
    analogWrite(led,intensity);
  }
  else if(inChar=='D'){ //command to decrease the light intensity
    intensity-=25;
    if (intensity<5) intensity=25;
    analogWrite(led,intensity);
  }
  inChar='1'; //to change reading inChar variable
}
```

4. Bluetooth Door Lock with Servo - v1

```
#include <Servo.h>
#include <SoftwareSerial.h> //library for the terminale mode with bluetooth

int redled=3;
int greenled=5;
Servo myServo;

char Data;
SoftwareSerial Bluetooth(0,1);

void sendData(String transmitData){ //function to send the data from the application to the module
  Bluetooth.println(transmitData);
}

void setup() {
  // put your setup code here, to run once:
  pinMode(redled,OUTPUT);
  pinMode(greenled,OUTPUT);
  myServo.attach(9);
  Bluetooth.begin(9600); //display data on the application
}

void loop() {
  // put your main code here, to run repeatedly:
  if(Bluetooth.available()){
    Data=Bluetooth.read();

    if(Data=='1'){
      digitalWrite(greenled,HIGH);
      digitalWrite(redled,LOW);
      myServo.write(90);
      sendData("green led turned ON and the door is opened");
    }

    if(Data=='2'){
      digitalWrite(redled,HIGH);
      digitalWrite(greenled,LOW);
      sendData("red led turned ON and the door is closed");
    }
  }
}
```

5. Bluetooth Door Lock with Servo - v2

```
#include <Servo.h>
#include <SoftwareSerial.h> //library for the terminale mode with bluetooth

int redled=3;
int greenled=5;
Servo myServo;

char Data;
SoftwareSerial Bluetooth(0,1);

void sendData(String transmitData){ //function to send the data from the application to the module
  Bluetooth.println(transmitData);
}

void setup() {
  // put your setup code here, to run once:
  pinMode(redled,OUTPUT);
  pinMode(greenled,OUTPUT);
  myServo.attach(9);
  Bluetooth.begin(9600); //display data on the application
}

void loop() {
  // put your main code here, to run repeatedly:
  if(Bluetooth.available()){
    Data=Bluetooth.read();

    if(Data=='1'){
      digitalWrite(greenled,HIGH);
      digitalWrite(redled,LOW);
      myServo.write(90);
      sendData("green led turned ON and the door is opened");
    }

    if(Data=='2'){
      digitalWrite(redled,HIGH);
      digitalWrite(greenled,LOW);
      myServo.write(0);
      sendData("red led turned ON and the door is closed");
    }
  }
}
```